GOVERNMENT OF KARNATAKA

DEPARTMENT OF SCHOOL EDUCATION (PRE UNIVERSITY)

Lesson Based Assessment UNIT TEST – 1

1st PU PHYSICS (33)

Unit 1 Chapters: Units and Measurements, Motion in a straight line

Maximum Marks: 30 **Total Time**: 1 Hour

General Instructions:

1. All parts (A to D) are compulsory.
2. For Part – A questions, only first written-answer will be considered for evaluation.
3. Answers without relevant diagram/figure/circuit wherever necessary will not carry any marks.
4. Direct answers to numerical problems without relevant formula and detailed solutions will not carry any marks.

Part A

I] Choose the correct answer from the given options **7 × 1 = 7**

1. The units that are used for fundamental physical quantities are called
a. System of units
b. Fundamental units
c. Derived units
d. All of these

2. Which of the following physical quantities has the same unit in CGS, FPS, MKS and SI system of unit?
a. Mass
b. Length
c. Time
d. energy

3. If the distance covered by a particle is zero, then its displacement is:
a. May or may not be zero
b. Cannot be zero
c. Must be zero
d. negative

4. The slope of the tangent drawn to position-time graph at any instant gives:
a. Instantaneous velocity
b. Instantaneous acceleration
c. Average velocity
d. Average speed

5. Stopping distance of a vehicle moving with uniform acceleration is directly proportional to:
a. Acceleration
b. Square of the acceleration
c. Initial velocity
d. Square of the initial velocity

6. When a body is thrown vertically upward, identify the WRONG statement among the following:
a. Velocity of the body at the highest point is zero.
b. Velocity of the body continuously changes.
c. Acceleration of the body continuously changes.
d. Acceleration of the body at the highest point is -9.8 m/s^2.

7. Statement I: A particle may be momentarily at rest and yet have non-zero acceleration
Statement II: This happens when acceleration of the object is equal to acceleration due to gravity.
a. Both the statements are true and II is the correct explanation of I.
b. Both the statements are true but II is not correct explanation of I.
c. Statement I is true but II is false.
d. Both the statements are false.

II] Fill in the Blanks by choosing the appropriate answer given in the bracket for ALL of the following questions. **3 × 1 = 3**

(*Scalar, Vector, Downwards, plane angle*)

8. _____ is a supplementary quantity.
9. Speed is a _____ quantity
10. A body moving under earth's gravitational field, the acceleration is always _____.

Part B

III] <u>Answer any TWO of the following questions. Each question carries 2 marks.</u> **2 × 2 = 4**

11. Write any two applications of dimensional analysis.

12. Write any two differences between instantaneous speed and instantaneous velocity.

13. A stone thrown vertically upward with velocity of 20 m/s. Calculate the maximum height reached by it. Take g = 10 m/s^2.

Part C

IV] <u>Answer any TWO of the following questions. Each question carries 3 marks.</u> **2 × 3 = 6**

14. Write the number of significant figures in (i) 12.2300, (ii) 25.6 × 10^8 and (iii) 2026

15. Derive the expression $v = v_0 + at$ using velocity-time graph.

16. Mention any three applications of velocity-time graph.

Part D

V] <u>Answer any ONE of the following questions. Each question carries 5 marks.</u> **1 × 5 = 5**

17. Centripetal force is a force experienced by a body in circular motion. It depends on the mass of the body (m), velocity of the body (v) and the radius of the circular path (r). Deduce an expression for centripetal force (F) in terms of m, v and r using dimensional analysis.

18. What is v – t graph? Derive the expression $v^2 = v_0^2 + 2ax$ using v – t graph.

VI] <u>Answer any ONE of the following questions. Each question carries 5 marks.</u> **1 × 5 = 5**

19. A car moving along a straight highway with a speed of 126 kmph is brought to a stop within a distance of 200m. What is the retardation of the car? How long does it take for the car to stop?

20. A body is thrown vertically up from the top of a building with a velocity of 10 m/s. It reaches round in 5s. Find the height of the building and the velocity with which the body reaches the ground. Take g = 10 m/s^2

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